

Research



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Multiple evolutionary transitions of reproductive strategies in a phylum of aquatic colonial invertebrates

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Parental care is considered crucial for the enhanced survival of offspring and evolutionary success of many metazoan groups. Most bryozoans incubate their young in brood chambers or intracoelomically. Based on the drastic morphological differences in incubation chambers across members of the order Cheilostomatida (class Gymnolaemata), multiple origins of incubation were predicted in this group. This hypothesis was tested by constructing a molecular phylogeny based on mitogenome data and nuclear rRNA genes 18S and 28S with the most complete sampling of taxa with various incubation devices to date. Ancestral character estimation suggested that distinct types of brood chambers evolved at least 10 times in Cheilostomatida. In *Eucratea loricata* and *Aetea* spp. brooding evolved unambiguously from a zygote-spawning ancestral state, as it probably did in *Tendra zostericola*, *Neocheilostomata*, and ‘*Carbasea*’ *indivisa*. In two further instances, brooders with different incubation chamber types, skeletal and non-skeletal, formed clades (*Scruparia* spp., *Leiosalpinx australis*) and (*Catenicula corbulifera* (*Steginoporella* spp. (*Labioporella* spp., *Thalamoporella californica*))), each also probably evolved from a zygote-spawning ancestral state. The modular nature of bryozoans probably contributed to the evolution of such a diverse array of embryonic incubation chambers, which included complex constructions made of polymorphic heterozoids, and maternal zooidal invaginations and outgrowths.

1. Introduction

Reproductive strategies in the animal kingdom are often described on an r–K continuum, with strict r-strategists producing many offspring while providing no parental care, and K-strategists producing fewer offspring and investing in higher levels of parental care [1,2]. Parental care is widespread across the animal kingdom, and diverse in its many forms [3]. In fact, examples of embryonic incubation can be found in approximately 80% of animal phyla, having evolved independently multiple (supposedly hundreds of) times, thus being one of the most outstanding examples of convergent evolution [4]. Although most animal species are oviparous, multiple examples of parental care across the animal kingdom indicate a strong adaptive force. Parental care is considered among the most important evolutionary novelties influencing the success of certain taxa via increased survival of their progeny [3,4].

One group in which various independent routes are hypothesized to have led to parental care of offspring, both in the form of (preparitive) viviparity and (postparitive) brooding, are bryozoans [4,5]. Bryozoa is a medium-sized phylum with approximately 6700 known extant species [6] of aquatic, almost entirely colonial, filter-feeding lophotrochozoans, the vast majority of which incubate their progeny [7,8]. In addition to a large diversity of colony morphologies [9], the modular nature of colonies allows for extraordinary zooidal polymorphism [10] and associated division of labour [11], including reproduction [5,8]. Embryonic incubation is present among extant representatives of all three bryozoan classes—in all Phylactolaemata (placental viviparity in internal embryo sacs), all Stenolaemata (intracoelomic placental viviparity with polyembryony) and most Gymnolaemata (both brooding and viviparity, sometimes placental). Structural and positional uniformity of the reproductive patterns and incubation devices indicate that embryonic incubation probably evolved just once in the former two classes, whereas it probably originated multiple times among gymnolaemates which show a wide array of reproductive traits [4,5,8]. As in other aquatic invertebrates, oviparity and planktotrophic development are considered an ancestral mode of reproduction in bryozoans [5,12].

The present study focuses on the gymnolaemate order Cheilostatmatida which is the most diverse bryozoan group today comprising 140 nominal families and more than 700 genera [6]. The earliest recorded fossil cheilostomes from the Late Jurassic [13] belong to the ‘malacostegans’, a paraphyletic group of non-brooders whose modern representatives produce long-lived planktotrophic cyphonautes larvae from numerous small eggs that are released via a tubular ciliated intertentacular organ (ITO) [14] (figure 1*a*). Although still common today, malacostegans never underwent a substantial radiation [15]. By contrast, the more advanced neocheilostomes, all of which incubate their offspring and produce short-lived, non-feeding (=endotrophic, i.e. lecithotrophic or placentotrophic) larvae [5,16–18], underwent a substantial radiation starting in the late Albian/early Cenomanian [15,19–21] and make up the vast majority of extant cheilostomes. Although suborder Neocheilostomina d’Hondt, 1985 [22] is not a currently accepted taxon name, ‘neocheilostome’ is a utilitarian term that will be used herein to refer to any cheilostome with an ovicell—a calcified brood chamber first found in the calloporid genus *Wilbertopora* from the late Albian [23].

In addition to ovicells, embryonic incubation in cheilostomes occurs in external and internal non-skeletal brood sacs and a wide range of calcified protective devices. Species of the genera *Aetea* (Aeteidae), *Eucratea* (Eucrateidae), *Leiosalpinx* (Leiosalpingidae) and ‘*Carbasea*’ *indivisa* (incertae sedis) incubate their embryos in external membranous brood sacs attached to the wall of the feeding module (autozoid) (figure 1*b*) [8,24,25]. The true nature of these sacs is unknown, and they were suggested to be either a product of the external zooidal wall [24], or sticky fertilization envelopes [5]. On the other hand, Steginoporellidae incubate their embryos inside fertile autozooids, in an internal brood sac—a non-calcified invagination of the body wall (figure 1*c*) [5,26]. In addition, there is a diversity of skeletal/calcified incubation chambers. The acanthostegal brood chambers in Tendridae are produced by periopodial spines that overarch the frontal membrane in the zoid distal to the maternal zoid [8,27] (figure 1*d*; electronic supplementary material, figure S1). Flattened kenozooids (polymorphic heterozooids) with flexible joints produce bi- to multi-valved brood chambers in Alysidiidae (termed synoecia in *Catenicula*) [28,29] (figure 1*e*), and kenozooids with a rigid base form bilobate brood chambers in Scrupariidae [5] (figure 1*f*). By contrast, most Thalamoporellidae have bilobate brood chambers formed by distal outgrowths of the frontal wall of the fertile (maternal) autozoid [5,30] (figure 1*g*). Apart from the viviparous Epistomiidae that incubate embryos intracoelomically [31], all neocheilostomes brood their progeny inside ovicells (figure 1*h*). Ovicells are formed of a protective outfold (ooecium) that is thought to have evolved from fused, overarching spines in a zoid distal to the maternal zoid [27,32]. Numerous modifications of ooecia are found across neocheilostome lineages. In many instances, ooecia are reduced to varying degrees, a process that was accompanied by the immersion of the brood cavity inside the zoid, resulting in the formation of non-calcified internal brood sacs [5,33].

Cheilostome brood chambers were suggested to have either had monophyletic [34,35] or polyphyletic origins [15,21]. Comparative analysis of their morphologies estimated seven or eight probable origins of brooding, i.e. in Aeteidae, Scrupariidae, Tendridae, Thalamoporellidae, Belluloporidae, ‘*Carbasea*’ *indivisa* (incertae sedis), Calloporidae and possibly Alysidiidae [5]. Previous molecular analyses showed that brooding cheilostome suborders Inovicellina (*Aetea*) and Scrupariina nest among non-brooding malacostegans, thus supporting a polyphyletic origin [36,37]. Meanwhile, Orr *et al.* [37] inferred five independent origins of brood incubation in the Cheilostatmatida. Here, three previously unsampled brooding taxa were added (*Catenicula corbulifera*, ‘*Carbasea*’ *indivisa*, *Tendra zostericola*) to create a molecular framework that includes the most exhaustive selection of cheilostome brooders to date, from five suborders (Inovicellina, Scrupariina, Tendrina, Thalamoporellina, Flustrina). Results from an ancestral character estimation (ACE) are discussed in light of morphological evidence. Our results corroborate earlier predictions of multiple evolutionary origins of diverse and complex brooding devices.

2. Material and methods

(a) Sample collection and taxon sampling

Samples were either collected from ponds or the intertidal zone, by scuba diving, snorkelling, by dredging, or by using a remotely

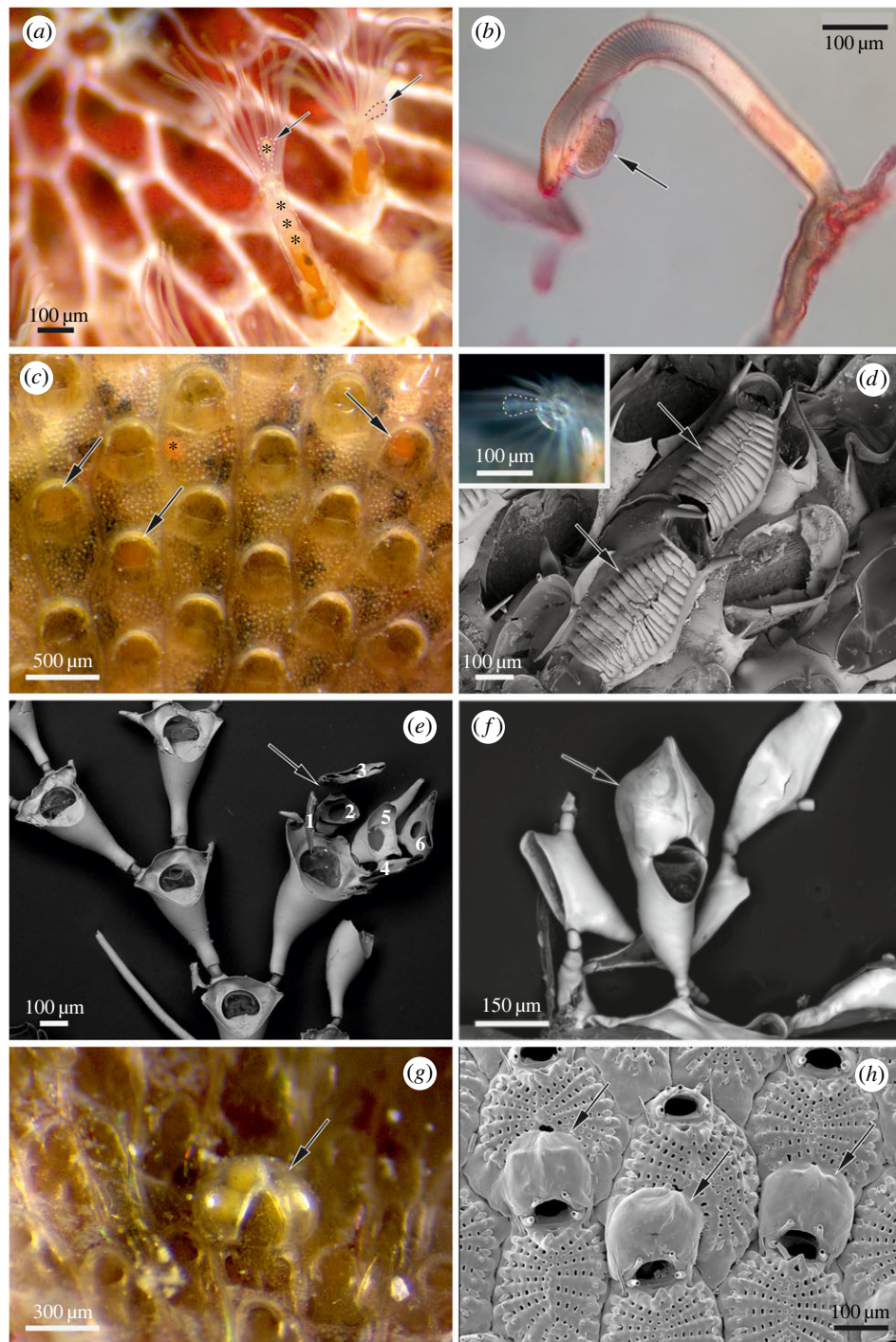


Figure 1. Reproductive structures of cheilostome bryozoans. (a) *Membranipora membranacea* with two expanded lophophores having an intertentacular organ (arrows and dotted lines); ripe zygotes shown by asterisks (photo by A. Gruhl, North Sea). (b) Autozoid of *Aetea* sp. with embryo inside external membranous sac (arrow) (NHMUK 1947.4.22.341). (c) *Steginoporella* sp. showing three fertile zooids with embryos (arrows) inside internal brood sacs; young oocyte shown by asterisk (photo by T. Schwaha, Great Barrier Reef, Australia). (d) *Tendra zostericola* showing two zooids with acanthostegal brood chambers (arrows) (inset: upper view of tentacle crown showing round mouth centrally and intertentacular organ to the left, dotted line) (NHMW 2022.VII.1). (e) *Catenicula corbulifera* showing fertile autozoid with synoecium (arrow) of six kenozooids (numbered 1–6) with cuticular joints (NHMUK 2023.6.22.6) (photo by Silviu Martha). (f) *Scruparia ambigua* showing fertile autozoid with bivalved brood chamber (arrow) made of two kenozooids (eastern Atlantic, northern Spain). (g) *Thalamoporella crenulata* showing fertile autozoid with bipartite brood chamber (arrow) containing three embryos (photo by T. Schwaha, Great Barrier Reef, Australia). (h) *Puellina egretta* showing three fertile zooids with ovicells (arrows) (Red Sea). NHMUK, Natural History Museum London; NHMW, Naturhistorisches Museum Wien.

operated vehicle (ROV), and preserved in 95–99% ethanol or RNA-later. Taxon sampling was focused on non-neocheilostomes with parental care: *Tendra zostericola* (acanthostegal brood chamber), *Scruparia* spp. (kenozooidal bilobate), *Steginoporellidae* (internal brood sac), *Thalamoporella californica* (maternal bilobate), *Catenicula corbulifera* (synoecium), *Leiosalpinx australis*, ‘*Carbasen*’ *indivisa*, *Aetea* spp. and *Eucratea loricata* (all external membranous brood sacs) (figure 1). Furthermore, members of the malacostegan

families Electridae, Membraniporidae and Sinoflustridae were included, which are known to perform zygote spawning without parental care. Published data for non-neocheilostomes from Orr *et al.* [37] were included to maximize taxon sampling. Also, neocheilostomes (with both ovicells and internal brood sacs) spanning 16 families were included. Non-cheilostome functional outgroups, as determined by Waeschenbach *et al.* [36] and Orr *et al.* [37] were included from all other major bryozoan lineages:

Pectinatella magnifica (class Phylactolaemata), *Exidmonea atlantica* and *Tubulipora flabellaris* (class Stenolaemata, order Cyclostomatida), *Anguinella palmata* and *Amathia verticillata* (class Gymnolaemata, order Ctenostomatida). Species which so far have never been put into a molecular phylogenetic context before are *Tendra zostericola*, *Leiosalpinx australis*, ‘*Carbasea*’ *indivisa*, *Catenicula corbulifera*, *Thalamoporella* sp., *Aetea* sp. (Seychelles), *Biflustra* new species (n. sp.), *Chaperiopsis* n. sp., and *Quadracellaria* cf. *caraibica*. For a full list of taxa with sampling localities and GenBank accession numbers, see electronic supplementary material, table S1.

(b) Morphological analysis

Specimens were imaged either using scanning electron microscopy (SEM) and/or light microscopy using stereo or compound microscopes (for detailed methodology and SEM images of morphological voucher specimens, see electronic supplementary material, S1 and figure S1).

(c) Wet laboratory methodology

The dataset for phylogenetic analysis was composed of partial and complete mitogenomes, as well as nuclear genes 18S and 28S ribosomal RNA (rRNA). Data for these genes were generated using the following methodologies: (i) Sanger sequencing of standard polymerase chain reaction (PCR) amplicons, (ii) Sanger sequencing of long-range mitogenome PCR amplicons, (iii) Illumina sequencing of long-range mitogenome PCR amplicons, and (iv) Illumina genome-skimming (for detailed sequencing, assembly and annotation methodologies, see electronic supplementary material, S1). Electronic supplementary material, table S1 provides an overview of newly generated and published data used in the analyses. Electronic supplementary material, S1 and tables S2–S4, outline taxon-specific PCR methodologies. For all methods, genomic DNA (gDNA) was extracted using the DNeasy Blood and Tissue kit (Qiagen). In most cases, small, clean colony fragments were used. For *Pectinatella magnifica* and *Oshurkovia littoralis*, dissected statoblasts and embryos were used, respectively.

(d) Phylogenetic analysis

Gene alignments were produced for mitochondrial (mt) protein-coding genes (PCGs), 12S and 16S rRNA genes, and for nuclear 18S and 28S rRNA genes. The newly generated data were combined with selected published data (see electronic supplementary material, table S1). Alignments were carried out in MAFFT [38]. Ambiguously aligned positions were identified using the Gblocks v. 0.91b web server (http://phylogeny.lirmm.fr/phylo.cgi/one_task.cgi?task_type=gblocks) [39,40] with the least stringent settings. The inclusion set provided by Gblocks was reversed into an exclusion set using Mesquite v. 3.51 [41]. This was done by pasting the included sites under ‘Begin Assumptions; Exset =’. In Mesquite, in ‘List of Characters’ view, all characters were highlighted, and ‘Reverse’ was selected under ‘Inclusion’. Best fitting models of nucleotide and amino acid substitution were determined for concatenations of (i) 12S + 16S rRNA genes (GTR + G4), (ii) 18S + 28S rRNA genes (GTR + G4) and (iii) amino acid translations of all mt PCGs (MTZOA + G4 + F) using the Akaike information criterion (AIC) implemented in ModelTest-NG v. 0.1.6 [42]. Each of the above-mentioned concatenations and the overall concatenated dataset, which was analysed as a mixed matrix (nuclear and mt rRNA genes as nucleotides and mt PCGs as amino acids), were analysed using Bayesian inference (BI) and maximum likelihood (ML). BI analyses involving amino acids were analysed in MrBayes5D v. 3.2.6 [43,44], whereas nucleotide-only analyses were performed in MrBayes v. 3.2.6 [43]. ML analyses were performed in RAxML HPC-PTHREADS-SSE3 v. 8.2.12 [45]. Because

MrBayes5D does not accommodate the +F model component, an analysis of the complete matrix was also performed in p4 [46] under the MTZOA + G + F model. Compositional heterogeneity across sequences was examined using posterior predictive simulations performed in p4; the test quantity was X^2 . BI analyses employing the NDCH-C2 model, which accommodates two composition vectors per data partition, were performed in p4. Robinson–Foulds distances were calculated in p4. In order to test the effect of missing data, BI and ML analyses of the complete matrix that excluded all terminals for which less than half of mitogenome genes were available, or which only had one of the two nuclear rRNA genes, were carried out in MrBayes5D and RAxML as outlined above. For analytical details, see electronic supplementary material, S1. The concatenated alignment with the exclusion set indicated, and the resultant trees, are available at <https://doi.org/10.5281/zenodo.8342054>.

(e) Ancestral character estimation

Ancestral character estimation (ACE) of reproductive strategies was carried out in *phytools* [47] and the *ace* function of the R package *ape* [48]. Taxa for which the reproductive strategy was unknown and duplicate sequences of the same species were both pruned off the phylogeny prior to analyses. The remaining taxa were coded as follows: outgroup—*placental viviparity in internal embryo sac* (Phylactolaemata), *tentacle sheath brooding* (Ctenostomatida representative *Amathia verticillata*), and *intracoeleomic placental viviparity with polyembryony* (Cyclostomatida); Cheilostomatida ingroup—*zygote spawning*, *internal brood sac*, *synoecium*, *maternal bilobate*, *kenozooidal bilobate*, *acanthostegal*, *external membranous brood sac I–IV*, and *neochelostome brood structure (ovicell or its derived internal brood sac)* (for taxon-specific coding, see electronic supplementary material, table S1; for images illustrating brood structures, figure 1). ACE was carried out using the equal-rate (ER) and symmetrical (SYM) models. An attempt to run the all-rates-different (ARD) model resulted in a ‘system is computationally singular’ error, which is probably due to the large number of variables. An AIC analysis of the two competing models favoured the ER (AIC = 129.68) over the SYM model (AIC = 278.22). For R code and analysis input files, see <https://doi.org/10.5281/zenodo.8342054>.

3. Results

(a) Novel molecular data

The newly generated data include 18 circularized mitogenomes. Twelve of those had the full complement of genes (13 PCGs, 22 transfer RNAs, two rRNA genes), while the remaining six had one missing transfer RNA each. Three further mitogenomes had the full complement of genes but could not be circularized. Six mitogenomes were partial, both in terms of gene content and circularization (for full details of mitogenome sizes, completeness, and gene order, see electronic supplementary material, table S5 and figure S2). The reasons for obtaining only partial mitogenomes were as follows: *Exidmonea atlantica* (AW004)—the region between *cox1* and *nad5* failed to amplify with long-range PCR; *Schizoporella dunkeri* (KC013)—the long-range PCR product between 12S rRNA and *cytb* could not be fully assembled and lacked the region between *trnW* and *nad4*; *Chaperiopsis* n. sp. (AW576)—the long-range PCR covering the region between *nad5* and *cox1* failed to amplify; assemblies of genome-skipped samples *Quadracellaria* cf. *caraibica* (AW2980), *Leiosalpinx australis* (AW1036) and *Tendra zostericola* (AW768) produced only partial mitogenome contigs. Furthermore, novel sequences were

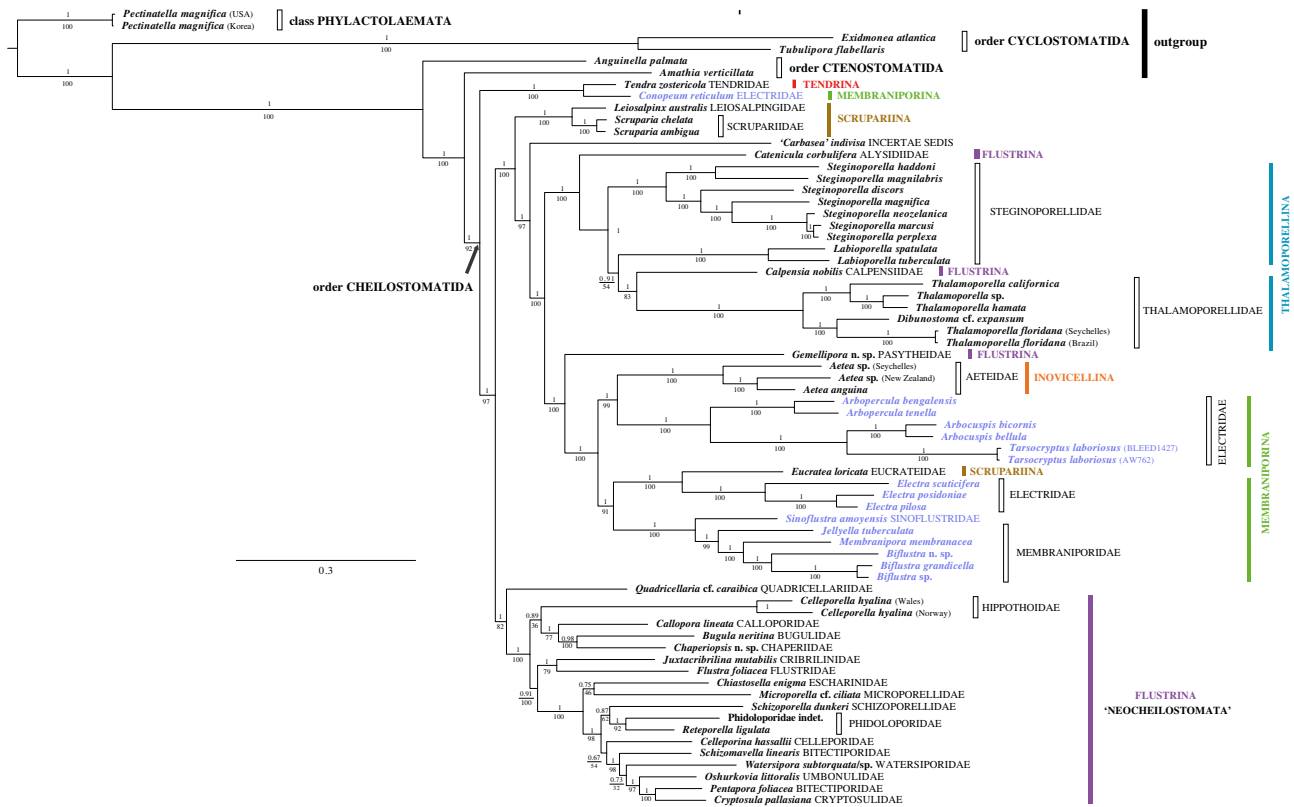


Figure 2. Bayesian phylogenetic analysis of the mixed concatenated alignment consisting of three partitions: (i) 13 mitochondrial protein-coding genes as amino acids, (ii) mitochondrial ribosomal RNA genes 12S + 16S, (iii) nuclear ribosomal RNA genes 18S + 28S. The analysis was performed in MrBayes5D v. 3.2.6 under the GTR + G model of nucleotide evolution (nucleotides) and the MTZOA + G model (amino acids). The analysis was run for 2.4 million generations; 1.5 million generations were discarded as burn-in. Posterior probabilities are given above the branches. Maximum-likelihood bootstrap support values generated using RAXML HPC-PTHREADS-SSE3 v. 8.2.12 over 100 replicates are given below the branches. The branch length-scale bar indicates number of substitutions per site. Higher level classification is given at the right-hand side. Cheilostomate suborders are indicated with different colours. Malacostegan-grade taxa are in light blue font. Whenever multiples of the same species/genus sp. are included, countries of origin or specimen codes are given.

generated for 18S rRNA ($n = 17$), 28S rRNA ($n = 15$), *cox1* ($n = 1$) and *cytb* ($n = 1$) (see electronic supplementary material, table S1).

(b) Molecular phylogeny

The alignment consisted of 68 taxa, 4111 amino acid positions (of which 719 were excluded) and 10 873 nucleotide positions (of which 4586 were excluded). This left 3392 and 6287 included amino acid and nucleotide positions, respectively. Posterior predictive simulations revealed significant compositional heterogeneity over the tree in the mt PCGs and 12S + 16S rRNA partitions (X^2 , $p < 0.0$). Although the use of the NDCH-C2 model increased the fit of the model to the data (see electronic supplementary material, table S7), convergence was only fair (average standard deviation of split frequencies: 0.02–0.04). Furthermore, the trees produced in the p4 analysis using the MTZOA + G + F model (electronic supplementary material, figure S3) and in the MrBayes5D analysis using the MTZOA + G model (figure 2) differed only by a Robinson–Foulds distance of four, and the posterior probabilities of the splits that differ were low (43 and 39); *Callopora lineata* + *Bugula neritina* + *Chaperiopsis* n. sp. were sister to *Celleporella hyalina* in the MrBayes5D MTZOA + G analysis, whereas in the NDCH-C2 analysis they were sister to a clade composed of Escharinidae, Microporellidae, Schizoporellidae, Phidoloporidae, Celleporidae, Bitectiporidae, Watersiporidae, Umbonulidae and Cryptosulidae. It therefore appears that using empirical composition (+F

model component) made little difference with this model. Furthermore, the BI and ML analyses that excluded taxa with significant amounts of missing data were identical to the MrBayes5D analysis (figure 2) concerning the taxa that remained included. Thus, the description of the molecular phylogeny is based on the MrBayes5D topology on which bootstrap values from the ML analysis were mapped (figure 2); the ML and MrBayes5D topologies were identical. For the ML topology of the overall alignment, the BI and ML trees of partitions 12S + 16S rRNA, 18S + 28S rRNA and mt PCGs, the p4 BI tree using the NDCH-C2 model, and the BI and ML trees that excluded taxa with significant amounts of missing data, see electronic supplementary material, figures S4–S13.

Sister taxa *Tendra zostericola* and *Conopeum reticulum* formed the sister group to all remaining cheilostomes (figure 2). Among the latter, a clade composed of *Leiosalpinx australis* and *Scruparia* spp. formed the sister group to the remaining non-neocheilostomes, plus four species currently classified as neocheilostomes, i.e. ‘*Carbassia*’ *indivisa*, *Catenicula corbulifera*, *Calpensia nobilis* and *Gemellipora* n. sp. ‘*Carbassia*’ *indivisa* was sister to the remaining non-neocheilostomes (plus *C. corbulifera*, *C. nobilis* and *Gemellipora* n. sp.), which formed two sister clades. In one of them, *Catenicula corbulifera* was sister to Steginoporellidae + Thalamoporellidae + *Calpensia nobilis*. Although members of the genus *Steginoporella* were monophyletic, the family Steginoporellidae was not, as members of the genus *Labioporella* formed the sister group to *Calpensia nobilis* + Thalamoporellidae. However, support

for this was poor (0.91 posterior probabilities (pp); 54% bootstrap support (bs)), and the NDCH-C2 analysis recovered the Steginoporellidae as monophyletic with moderate support (0.95 pp; see electronic supplementary material, figure S11). Among the Thalamoporellidae, *Dibunostoma* cf. *expansum* nested within *Thalamoporella* species. In the second clade, the formal neocheilostome *Gemellipora* n. sp. was the sister to a clade composed of Inovicellina, Membraniporina and Scrupariina. This clade was split into two subclades. One of them consisted of *Aetea* spp. and electrid genera *Arbocuspis*, *Tarsocryptus* and *Arbopercula*. In the other subclade *Eucratea loricata* formed the sister group to *Electra* spp., whereas *Sinoflustra amoyensis* was sister to Membraniporidae, represented by genera *Jellyella*, *Membranipora* and *Biflustra* (figure 2).

Monophyly of the neocheilostomes (except *Catenicula corbulifera*, *Calpensia nobilis*, and *Gemellipora* n. sp.) was fully supported, although the support of some nodes was poor. The interrelationships within this clade are not the focus of our study and are not discussed here.

(c) Ancestral character estimation

The ancestral reproductive mode of Bryozoa was estimated as *viviparity in internal embryo sac* with 0.71 empirical Bayesian posterior probability (EBPP) support (figure 3) (for node- and character state-specific EBPP values, see electronic supplementary material, table S6). This mode is ubiquitous among phylactolaemates, here represented by *Pectinatella magnifica*. The ancestral states from which the viviparous cyclostome and brooding ctenostome outgroups evolved could not be estimated without ambiguity. The support for the common ancestor of cyclostomes and gymnolaemates to have reproduced via *viviparity in internal embryo sacs* was 0.48 EBPP (node 2), whereas the support for the common ancestor of the gymnolaemates to have performed *zygote spawning* was 0.56 EBPP (node 4).

The ancestral reproductive mode of Cheilostatmatida was estimated as *zygote spawning* with 0.62 EBPP support (node 5). In fact, all remaining backbone nodes were reconstructed as mostly *zygote spawning* with EBPP values ranging from 0.61 to 0.71 (nodes 7, 25, 28, 29). There are two instances in which brood incubation evolved unambiguously and independently from a *zygote-spawning* condition (0.99 EBPP; nodes 41, 49). They are *external membranous sac III* and *IV*, found in *Aetea* spp. and *Eucratea loricata*, respectively. There is also some evidence that the *acanthostegal brood chambers* found in *Tendra zostericola* evolved from a *zygote-spawning* condition (0.72 EBPP; node 6). The *maternal bilobate* brood chamber in *Thalamoporella californica* was estimated to have evolved from the *internal brood sac* condition in Steginoporellidae (0.97 EBPP; node 38). In turn, there is reasonable support that the *internal brood sac*, the *synoecium* in *Catenicula corbulifera* and the *external membranous sac II* in '*Carbasea*' *indivisa* may have evolved from a *zygote-spawning* ancestor (0.67 EBPP, node 28). Both the *kenozooidal bilobate* brood chamber in *Scruparia* spp. and *external membranous sac I* in *Leiosalpinx australis* may also have evolved from a *zygote-spawning* condition (0.64 EBPP, node 25). However, the reproductive mode of their immediate ancestor was ambiguous (0.47 EBPP *kenozooidal bilobate*; 0.34 EBPP *external membranous sac I*, node 26). The support for the common ancestor of non-neocheilostomes (to the exclusion of *Conopeum reticulum* and *Tendra zostericola*)

and neocheilostomes to have been a *zygote spawner* was 0.61 EBPP (node 7).

4. Discussion

(a) Phylogenetic distribution of reproductive modes

Oviparity and planktotrophic development are considered ancestral modes of reproduction among aquatic invertebrates, with *zygote spawning* having evolved from it after acquisition of internal fertilization (reviewed in [5,12]). Our ACE analysis estimated, with reasonable support (EBPP = 0.71), that *viviparity in internal embryo sacs* and *placentalotrophy*, as found in phylactolaemates, is the plesiomorphic mode of reproduction in bryozoans. However, this mode has been inferred entirely from extant taxa. It is also considered as highly derived in other invertebrates [4]. Since phylactolaemates lack any mineralized skeleton, their sparse fossil record consists of occasionally preserved statoblasts, the asexually produced resting stage. The earliest fossil evidence for this group comes from the Upper Triassic [49], possibly even the Permian [50], but molecular clock estimates place the divergence of Phylactolaemata from Myolaemata (Stenolaemata + Gymnolaemata) as early as the Ediacaran [51]. Most phylactolaemates produce numerous small oligolecithal eggs [52], a condition ubiquitous for *zygote-spawning* gymnolaemates having planktotrophic larvae. Thus, although our ACE analysis based on living taxa estimated *viviparity in internal embryo sacs* as plesiomorphic, it is conceivable that a shift from plesiomorphic *zygote spawning* in ancestral phylactolaemates to advanced viviparous embryonic incubation might have taken place during their early evolutionary history and in conjunction with their hypothesized transition from marine to freshwater environment, which would have been accompanied by numerous changes in morphology, physiology, reproduction and development [7,8,53].

For cheilostomes, the ACE analysis provided moderate support (EBPP = 0.62, node 5) for *zygote spawning* to be their ancestral mode of reproduction, and our results support the notion that the various brood incubation modes in this group have evolved repeatedly from this ancestral state (figure 3). The *zygote spawning* species *Conopeum reticulum* (Electridae), together with the brooding species *Tendra zostericola* (Tendridae) formed the sister group to the remaining cheilostomes. *Conopeum reticulum* was also recovered in this position by Waeschenbach *et al.* [36], whereas in Orr *et al.* [37] *Conopeum* spp. formed the sister group to the neocheilostomes, albeit with only 61% bootstrap support. The phylogenetic position of *Conopeum* probably led to the '*zygote-spawning*' condition to be pulled towards the ancestral node of Cheilostatmatida (node 5). In *Tendra zostericola*, which broods endotrophic larvae in a space between frontal membrane and periopodial spines, the simple morphology of its autozooids is reminiscent of malacostegan-grade cheilostomes, especially Electridae, and this species was sometimes classified as either *Electra* or *Membranipora* [28,35,54,55]. Thus, the close association between *T. zostericola* and the nominal electrid *C. reticulum* is not too surprising. Furthermore, *T. zostericola* incubates 4–10 embryos simultaneously, whereas *C. reticulum* typically expels 5–9 zygotes (in [5]). Also, their oocyte sizes are similar: 110 × 80 µm in *C. reticulum* [56] and 70 µm in *T. zostericola* [55]. Thus, despite a shift towards brooding and lecithotrophy in *T. zostericola*, the two species share

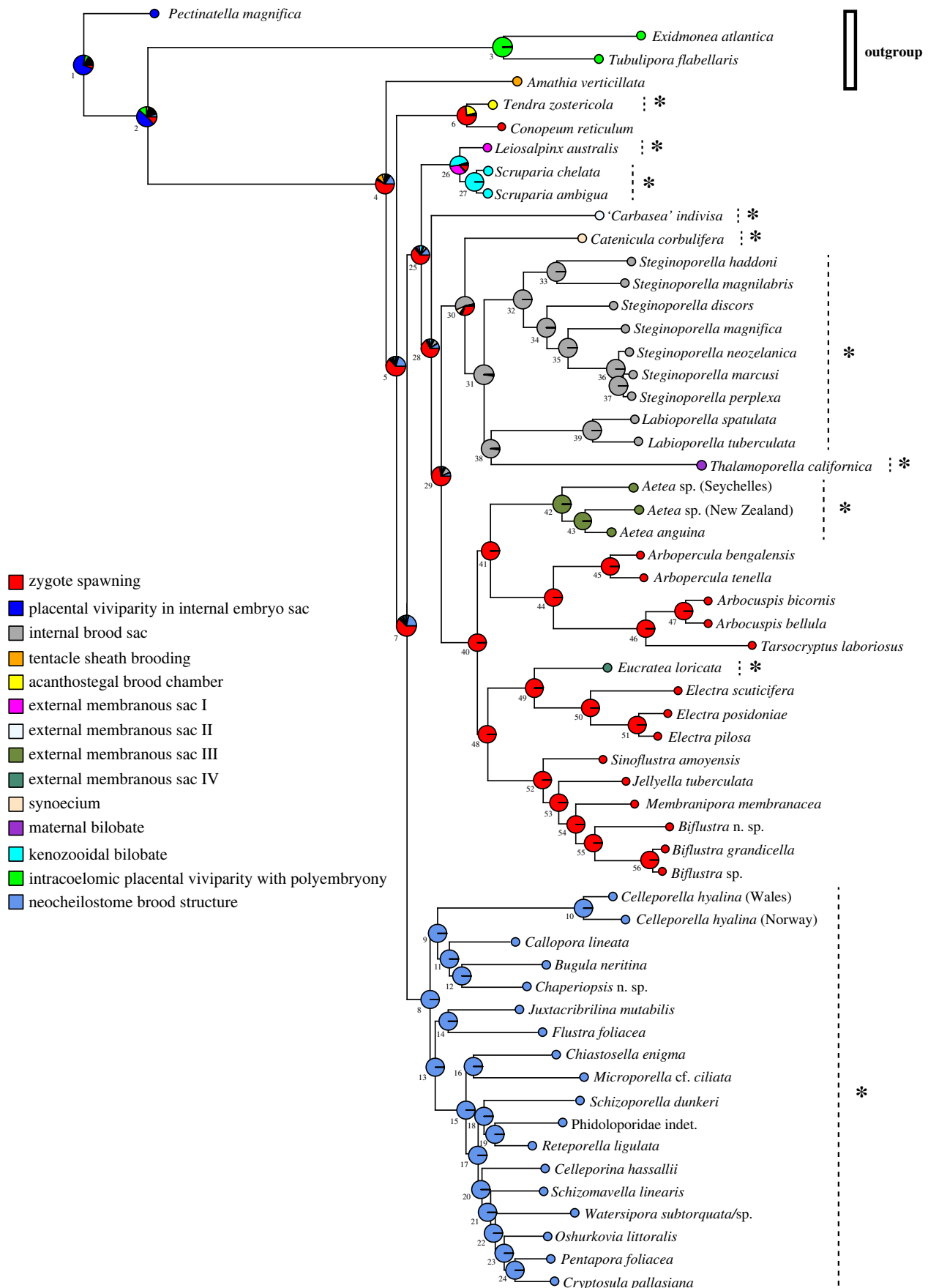


Figure 3. Ancestral character estimation of reproductive modes performed in *phytools* using the *ace* function and the equal-rates (ER) model implemented in the R package *ape*. The empirical Bayesian posterior probabilities (EBPP) which indicate the probability of occurrence of character states is given as pie charts at nodes. For EBPP values corresponding to the numbered nodes, see electronic supplementary material, table S6. Asterisks and associated dashed lines indicate the distribution of distinct brood chamber types.

morphological and reproductive traits. In view of these facts, the definition of the suborder Tendrina Ostrovsky, 2013 should consider the polyphyly of the Electridae, which was

recovered here as well as by Orr *et al.* [37], and be expanded to include *Conopeum*. Lastly, the distant phylogenetic position of *T. zostericola* from neocheilostomes, and *Juxtacribrilina*

mutabilis in particular, refutes the hypothesis by Braiko [55] and Santagata & Banta [35] that the acanthostegal brood chamber is a primitive form of cribrimorph ovicells. Instead, the alternative hypothesis that the acanthostegal brood chamber in *Tendra* and neocheilostomes ovicells are non-homologous [27] is supported by our results.

As in Orr *et al.* [37], Scrupariina were polyphyletic. Our inclusion of *Leiosalpinx australis* demonstrated the monophyly of the Leiosalpingidae and Scrupariidae, which together formed the sister group to the rest of the non-neocheilostomes plus '*Carbasea*' *indivisa*, *Catenicula corbulifera*, *Gemellipora* n. sp. and *Calpensia nobilis*. Concerning the Eucrateidae, as previously recovered by Orr *et al.* [37], *Eucratea loricata* was sister to *Electra* spp., and together they formed the sister group to Sinoflustridae and Membraniporidae (figure 2). Regarding reproduction, according to our ACE analysis, the external membranous sac IV in *E. loricata*, which is attached distally to the zooidal orifice and in which single embryos are incubated [57], has unambiguously evolved from a zygote-spawning ancestral state (node 49; figure 3). Interestingly, *Aetea* spp., which produce similar external membranous brood sacs III, nested in the sister clade to the above-mentioned clade and also unambiguously evolved parental care from a zygote-spawning ancestral state (node 41; figure 3). Relative to *Eucratea*, the membranous brood sac is on the erect portion of the zooid in *Aetea* (e.g. [58]; figure 1b), although the precise attachment position may vary (see [5]). These sacs sometimes contain up to two larvae [24]. *Aetea* spp. formed the sister group to electrids *Arbopercula* spp. + *Tarsocryptus laboriosus* + *Arbocuspis* spp., a topology found also by Orr *et al.* [37]. Thus, the topology indicates that these similar external membranous brood sacs found in eucrateids and aeteids probably evolved independently from the same ancestral zygote-spawning ancestor (node 40; figure 3).

Despite the similarity of the external membranous brood sac found on the frontal membrane of the zooids of *Leiosalpinx australis* [25] to those found in *Eucratea*, the two taxa are unrelated and their respective incubation structures must have arisen independently too (figures 2 and 3). Interestingly, the closely related Scrupariidae (including the genus *Brettiopsis* that was not sampled here) have evolved an entirely unique terminal, helmet-like bipartite brood chamber, produced by two kenozooids budded from the maternal autozooid (figure 1f) [5]. Although both incubation modes in Scrupariidae and Leiosalpingidae possibly evolved independently from an ancestral zygote-spawning state [5] (node 25; figure 3), the character state of their most common recent ancestor was equivocal (node 26; figure 3). The notion that the external membranous brood sacs in aeteids might indicate a close relationship with ctenostomes that have similar incubation structures, e.g. *Pottsiella* [59], is not supported by our phylogeny. Furthermore, Scrupariina, as defined by Ostrovsky [5], ought to be revised to the possible exclusion of the Eucrateidae.

A taxon with yet another kind of external membranous sac (external membranous sac II) is '*Carbasea*' *indivisa*. It is currently classified as a neocheilostome, although it has been recognized as a flustrid homeomorph (i.e. having erect, lobate and lightly calcified colonies with elongated zooids as in most Flustridae). This taxon was sister to Thalamoporellina, Inovicellina, Membraniporina (minus *Conopeum reticulum*), *Catenicula corbulifera*, *Calpensia nobilis*, *Gemellipora* n. sp. and *Eucratea loricata* (figure 2). Described as being formed by outgrowths of the

tentacle sheath at the distal edge of the operculum, these brood sacs, of which there can be three to seven at any one time [60], could be unique among bryozoans. However, these sacs were never studied anatomically, and both the description and illustration by Stach [60] are speculative. Multiple embryos attached to either cystid or polypide are well-known among ctenostomes (reviewed in [5]). It is possible that brood-sac formation in '*C.*' *indivisa* is analogous to that in the above-mentioned brooders (*Aetea*, *Eucratea*, *Leiosalpinx*) and various ctenostomes, and may represent emergence of the fertilization envelope (reviewed in [5]). Thus, our ACE analysis provided support for the hypothesis that these simple incubation structures have had multiple origins from a zygote-spawning ancestral state in cheilostome bryozoans (node 28; figure 3).

The kenozooidal brood chambers of Alysidiidae represent yet another type of incubation device. Although superficially similar to the rigid kenozooids forming brood chambers in Scrupariidae, each kenozooidal valve has a cuticular base that permits it to bend outwards in the 'bivalved' brood chamber of *Alysidium parasiticum* [28] and in the more complex *synoecia* in the confamilial genus *Catenicula*, which consist of six kenozooids interconnected by elastic cuticular joints (figure 1e) [29]. No *Alysidium* samples could be obtained for DNA sequencing, but this taxon should be included in future studies to test the monophyly of the family.

Catenicula corbulifera formed the sister group to Thalamoporellina and *Calpensia nobilis* (figure 2). Yet, its *synoecium* and the internal brood sac II found in Steginoporellidae (Thalamoporellina) are entirely different types of incubation chambers, both of which may have arisen from a zygote-spawning ancestral state (node 29; figure 3) via employment of kenozooids or acquisition of vestibular brooding [35]. Furthermore, based on our analysis, there is strong support that the maternal bilobate incubation chamber found in many species of *Thalamoporella*, including *T. californica*, which is a prominent bipartite semispherical structure produced by distal protuberances of the frontal wall of the maternal zooid (figure 1g) [5,30], may have evolved from the internal brood sac II of Steginoporellidae (node 31; figure 3). This contradicts a hypothesis developed for neocheilostomes that internal brood sacs evolved by immersion of the brood cavity of initially prominent ovicells into the colony [5,33]. Considering that there is topological conflict regarding the position of *Labioporella* spp. and the monophyly of Steginoporellidae (figure 2; electronic supplementary material, figures S3–S11; [37]), this result requires further investigation.

Also, our results support a previous assumption by Harmer [26] that Thalamoporellidae and Steginoporellidae are closely related. Furthermore, the position of the Thalamoporellidae among the non-neocheilostomes confirms the previous hypothesis that their ovicell-like brood chambers evolved independently from neocheilostome ovicells [5,26]. Notably, the presence of an ITO in *Thalamoporella floridana*, combined with the lack of evidence for brood chambers (L Vieira, since 2005, personal observations), indicates that such species of *Thalamoporella* are probably zygote spawners (see also §4(c)). If this is the case, the estimated reconstruction of reproductive modes at nodes 30 and 31 (figure 3) would probably include a higher proportion of 'zygote-spawning' and could infer two independent transitions to brooding within Thalamoporellina, among Steginoporellidae and Thalamoporellidae. Finally, the presumed homology of

'two-valved' brood chambers, as found in *Alysidium*, *Thalamoporella* and *Scruparia*, as suggested by Hyman [61], is not supported by phylogenetic analyses, which resolved Scrupariidae and Thalamoporellidae as non-monophyletic.

Neocheilostomes formed the sister group to the non-neocheilostomes minus *Tendra zostericola* and *Conopeum reticulum* (figure 2). Although ovicells in this group have evolved into a range of morphological types [5], they are thought to have had a single origin, with the earliest evidence coming from the calloporid *Wilbertopora mutabilis* from the late Albian/Early Cretaceous [23]. These early ovicells are hypothesized to have formed by the overarching articulated periopodial spines belonging to the zooid distal to the maternal zooid [32,62]. Our ACE analysis infers a separate origin of the neocheilostome brood structure (ovicells together with their derived internal brood sacs) from a possible zygote-spawning ancestral state (node 7; figure 3), which supports a hypothesis based on the fossil record that considers zygote-spawning malacostegans (Membraniporina) as possible ancestors of ovicellate cheilostomes [27,32].

(b) Traits in brooders that point to a zygote-spawning evolutionary past

There are several traits found in some of our studied species that point to a zygote-spawning evolutionary past. Most known malacostegans possess an ITO via which they accept alien sperm and release fertilized eggs (figure 1a) [8,14]. This organ is absent in neocheilostomes, but present in *Tendra zostericola* (figure 1d, inset) [54,55] and *Thalamoporella evelinae* (not sampled here) [30], which use it for oviposition of fertilized oocytes to the brood chamber. The presence of this organ clearly points to the origination of these brooders from zygote-spawning, malacostegan-like ancestors. Furthermore, *Tendra*, *Steginoporella*, *Thalamoporella*, *Scruparia* and *Aetea* simultaneously brood two to several embryos in the same brood chamber (figure 1g; electronic supplementary material, figure S1), a condition that is only known in the early diverging neocheilostome genera *Macropora* and *Monoporella* [37] and two internal brooders, *Oshurkovia littoralis* and *Arctonula arctica* [5]. The release of eggs in cohorts is characteristic for zygote spawners, yet the brooder *Tendra zostericola* also follows this pattern. Regarding *Leiosalpinx australis* and 'Carbasea' *indivisa*, 1–2 [25] and 3–7 embryos [60] are brooded simultaneously. 'Twinned larvae' were also detected in one species of *Steginoporella* [58]. Multiple embryonic incubation is considered a rudiment of the primitive reproductive pattern that includes production and release of numerous eggs in broadcasters and zygote spawners. Moreover, the endotrophic larva of *Tendra* reportedly has a rudimentary gut that is probably homologous with the gut of planktotrophic cyphonautes larvae of zygote-spawning malacostegans [63–65]; such larvae with a rudimentary gut are also present in some ctenostomes but are absent in neocheilostomes [16,53].

(c) Taxa with unknown or uncertain reproductive modes

There are several taxa in our dataset for which the reproductive mode is unknown or uncertain, and which were excluded from the ACE analysis accordingly. The ctenostome *Anguinella palmata* is an internal brooder although details are

currently unknown (T Schwaha 2020, personal communication). No brood chambers have been observed in *Calpensia nobilis* [66] and brooding is assumed to take place in an internal brood sac [5], although this has not been directly observed. As some species of Thalamoporellidae are probably zygote spawners (see above), *Calpensia* could potentially have this reproductive pattern, too.

Among the Thalamoporellidae, the brooding reproductive mode is only known for one of the species in our dataset, *T. californica*. The available material of *Thalamoporella* sp. did not have any brood chambers, and the condition in *T. hamata* is unknown. *Thalamoporella floridana* has an ITO, and incubation chambers were never observed (L Vieira, since 2005, personal observations), which suggests it might be a zygote spawner. Regarding *Dibunostoma* cf. *expansum*, no embryonic incubation has ever been observed in *D. expansum* (D Gordon 2023, personal observations). Furthermore, no brood chambers have been described in congeners *D. purii* Cheetham, 1963 [67] and *D. gracile* (Maplestone, 1900) [68]. No incubation chambers were present in our material of *Quadracellaria* cf. *caraibica* and none were ever reported in the descriptions of *Q. caraibica* [69]. Regarding *Gemellipora* n. sp., the absence of brood chambers in this genus [58] has been interpreted as likely evidence for incubation in internal brood sacs [5], but its placement in our molecular phylogeny suggests that it could be also a zygote spawner. The absence of external incubation chambers in the above-mentioned taxa requires further anatomical study to clarify whether incubation is indeed performed internally. Furthermore, *Bellulopora bellula*, which is currently classified in its own suborder Belluloporina, was not included in the present study due to the unavailability of tissue. This species produces unique, perforated, oocia-like skeletal chambers made of kenozooidal costae [27], which probably represent another independently evolved incubating device. Finally, intracoelomic viviparity in the neocheilostome Epistomiidae probably evolved from embryonic brooding in ovicells [5].

(d) Evolutionary significance of parental care

Different modes of brood incubation are known from 28 of the 33–34 recognized animal phyla, which in many cases evolved multiple times within each phylum [4]. Current estimates show that viviparity (preparative incubation) alone originated independently more than 150 times among vertebrates [70]. The independent origins of brooding or viviparity were never evaluated for invertebrates, but considering their enormous taxonomic diversity, it probably exceeds that in vertebrates. To mention a few examples, embryonic incubation evolved at least 25 times among hexacorals [71,72], 18 times among echinoderms [4], and at least 61 times among dipteran insects [73].

Regarding cheilostome bryozoans, Orr *et al.* [37] estimated five separate origins of brood incubation from an ancestral zygote-spawning state. However, the finer granularity of our coding and the addition of crucial, previously unsampled taxa (*Catenicula corbulifera*, 'Carbasea' *indivisa*, *Tendra zostericola*), raised this number to at least 10 separate origins (nine origins from an ancestral zygote-spawning state, while brooding *Thalamoporella* species could have evolved from taxa that perform internal brooding) (figure 3). *Bellulopora* and Epistomiidae represent two more potential examples.

This raises the question: what are the evolutionary drivers that favour shifts from oviparity, which still prevails in many animal lineages, to embryonic incubation [2]? Mathematical models [1,74], comparing parental productivity and youth survival rate in marine invertebrates, predict that a strategy with small eggs and a large number of offspring (e.g. broadcasting/zygote spawning) combined with planktotrophy should be equally stable, in evolutionary terms, as the combination of endotrophy with larger eggs and a smaller number of offspring. Among aquatic invertebrates, the risk from predation faced by long-lived planktonic larvae is compensated by the numerous offspring. Shifts from planktotrophic to endotrophic larvae with a shorter free-swimming period are often associated with a transition from production of many small to fewer larger eggs, i.e. reduction in progeny number. This reduction may be compensated by either faster development or by acquisition of embryonic incubation or both [5,12,75]. Such transitions and their evolutionary advantage(s) are extensively discussed in the literature (reviewed in [3]).

Taylor [21] hypothesized that the origin of brooded, short-lived, endotrophic larvae could have been a trigger for the diversification of cheilostomes because their truncated planktonic life would have resulted in poorer dispersal potential and, thus, reduced gene flow among distant populations, thereby facilitating speciation [76]. It was also suggested that endotrophic larvae could be advantageous at greater ocean depths where algal planktonic food is scarce [5]. One argument in favour of these hypotheses is that zygote-spawning malacostegans were (and still are) much less diverse and mostly occupy shallower habitats, while brooding neocheilostomes, which underwent a substantial radiation starting in the late Albian/early Cenomanian [15,19–21], still make up the vast majority of cheilostomes for all substrates and depths today.

On the other hand, the evolutionary importance of corresponding multiple transitions to incubation and endotrophic larval type is currently obscure because most of the non-neocheilostome brooding taxa, which are typically weakly calcified and lack any biomineralized frontal wall or cryptocyst, are relatively species poor. An exception is the species-rich *Thalamoporellina*, which includes known brooders, all of which possess a well-developed protective cryptocyst. While Gordon & Voigt [77] agreed with the idea by Taylor [21], they also highlighted the potential importance of the origins of calcified frontal walls for the sustained increase of species numbers into the Late Eocene, including recovery from the K-Pg extinction. Also, neocheilostomes are characterized by a high degree of zooidal polymorphism, which might also have contributed to their success [11]. Furthermore, Moharrek *et al.* [19] found that global cheilostome genus origination rates were positively correlated with sea surface temperature. Thus, cheilostome macroevolutionary dynamics are probably affected by a multitude of factors.

Moreover, the evolution of embryonic incubation among bryozoans was often accompanied by acquisition of matrotrophy (extraembryonic nutrition), an important evolutionary novelty found in both vertebrates and invertebrates (reviewed in [4,70]). Matrotrophy probably accelerates the rate of larval production, which is particularly important for ephemeral species which have limited time to produce their offspring [5,17,18]. In Cyclostomatida, the evolution of placentation resulted in the origin of polyembryony—a continuous production of clonal larvae—which probably promoted their long-lasting evolutionary success since the Late Triassic [78].

Lastly, the modular nature of bryozoans, and the high level of polymorphism found in the Cheilostomatida in particular, is probably an important prerequisite for the evolution of the diverse array of brood chamber types as presented here. Whereas all phylactolaemates, cyclostomes and some gymno-laemates use the entire autozoid or their modified or non-modified parts (body cavity, body-wall invaginations and outgrowths, vestibulum and tentacle sheath) as incubation chambers, the vast majority of cheilostomes construct ‘marsupial’ devices of zooidal polymorphs—kenozooids (*Catenicula*, *Alysidium*, *Scruparia*, *Brettiopsis*, *Bellulopora*) and spines (*Tendra*, neocheilostomes), thus showing multiple examples of exaptation [78]. Wall outgrowths (modified spines?) forming bipartite brood chambers in *Thalamoporella* and external membrane brood sacs represent the third and fourth routes to offspring incubation, respectively.

5. Conclusion

Our discovery of multiple independent origins of brooding in Cheilostomatida exemplifies how zooidal modules and their components can be adapted to produce an extraordinary diversity of chambers for the incubation of endotrophic larvae, most likely from a non-brooding, planktotrophic ancestral state. It is hypothesized that brood incubation in this clade repeatedly originated in concert with the evolution of an endotrophic larval stage that, in turn, resulted from a shift from oligolecithal to macrolecithal oogenesis. In contrast to other marine invertebrates, no endotrophic larvae are known in bryozoan zygote spawners. It is therefore hypothesized that incubation and endotrophy evolved in concert, or embryonic incubation anticipated endotrophy.

The large diversity of bryozoan reproductive patterns indicates high adaptive plasticity in this group. The multiple shifts between two major reproductive strategies that affected oogenesis, embryonic incubation and larval types are key innovations of cheilostome bryozoans that together with other novelties probably promoted their evolutionary success during the last 100 Myr.

Ethics. This work did not require ethical approval from a human subject or animal welfare committee.

Data accessibility. The data are provided in electronic supplementary material [79].

Declaration of AI use. We have not used AI-assisted technologies in creating this article.

Authors' contributions. H.E.G.: data curation, formal analysis, writing—original draft, writing—review and editing; A.N.O.: conceptualization, investigation, resources, writing—original draft, writing—review and editing; H.L.J.: data curation, formal analysis, writing—review and editing; L.M.V.: investigation, resources, writing—review and editing; D.P.G.: investigation, resources, writing—review and editing; P.G.F.: formal analysis, writing—review and editing; O.N.K.: resources, writing—review and editing; A.M.S.: resources, writing—review and editing; B.B.: resources, writing—review and editing; J.S.P.: resources, writing—review and editing; J.S.: resources, writing—review and editing; W.K.F.: resources, writing—review and editing; K.J.T.: resources, writing—review and editing; A.W.: conceptualization, data curation, formal analysis, funding acquisition, methodology, project administration, resources, supervision, writing—original draft, writing—review and editing.

All authors gave final approval for publication and agreed to be held accountable for the work performed therein.

Conflict of interest declaration. We declare we have no competing interests.

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